Tangents

1 $f'(2) = 17$

2

a $\dfrac{dy}{dx} = x^2 + x^5 + 2$

b $-16\,756$

3 $f'(4) = -1$

4

a $x = -5, x = 2$

b $f'(2) = 7$

5

a $\dfrac{dy}{dx} = 8x - 5$

b $x = \dfrac{5}{8}$

6

a $f'(2) = -12$

b $y = -12x + 16$

7

a $f'\left(\dfrac{1}{9}\right) = \dfrac{15}{2}$

b $y = \dfrac{15x}{2} + \dfrac{5}{6}$

8

a $f'(4) = \dfrac{3}{2}$

b $y = \dfrac{3x}{2} + 6$

9

a $f'(-2) = -\dfrac{3}{8}$

b $-\dfrac{1}{4}$

c $y = -\dfrac{3x}{8} - 1$

10

a 8

b $y = 4x + 4$

11

a To find the equation of the tangent line at $x = 2$, we need to find the gradient of the tangent line at that point, which is given by $f'(2)$. We also need to find the y-coordinate of the point at $x = 2$. We can then substitute the known gradient and point into the equation of a line ($y = mx + c$) to find the value of c.

b $y = 36x - 48$

12

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a $\frac{5}{3}$

b $y = -\frac{4x}{3} + \frac{1}{3}$

13

a $g'(x) = 20x^4 - 12x^3 + 6x^2 - 7x^{-3}$

b $y = 7x - \frac{1}{2}$

14 $y = 5x - 4$

15 $y = 7x - 19$

16 $(15, 2025), (-11, -2057)$

17 $(2, -16), (-4, 62)$

18 $(9, -486)$

19 $x = \frac{3}{4}, x = -2$

20 $(2, 4)$

21 $(2, 8)$

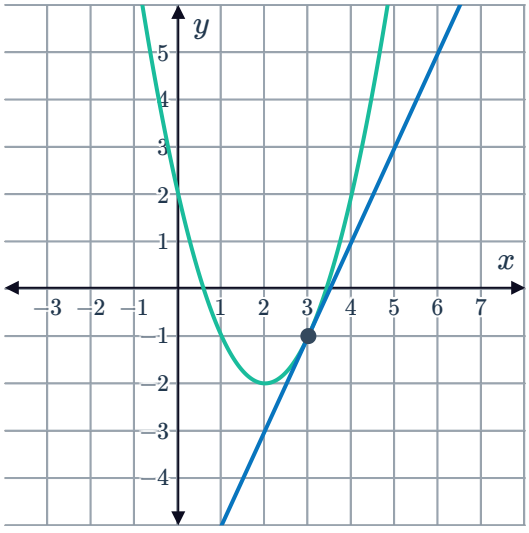
22 $y = -13$

23

a 2

b $(2, -2)$

c

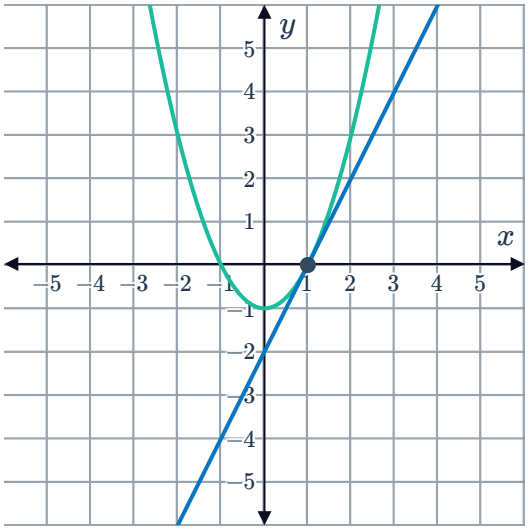


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24

a 2

b



25

a $x = \frac{1}{2}, x = \frac{1}{6}$

b At $x = \frac{1}{2}$ the tangent is $y = -\frac{1}{4}x$ and at $x = \frac{1}{6}$, the tangent is $y = -\frac{x}{4} + \frac{1}{54}$.

26 $k = 40$

27

a $a = 900$

b $b = 90$

28

a $\frac{dy}{dx} = 2x + b$

b -5

c $b = -23$

d $c = 79$

29

a $b = 14 - 3a$
$b = 11 - 2a$

b $a = 3, b = 5$

30

a $c = 0, d = -28$

b $b = -2a + 7$
$b = 9 - 3a$

c $a = 2, b = 3$

31

a $m = 8, m = 4$

b $y = 8x - 22$

c $y = 4x - 10$

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Normals

32

a $\dfrac{dy}{dx} = 50x^4 - 8x^3 + 24x^2 - 12x + 4$ **b** 4018

c $\dfrac{-1}{4018}$

33

a 2 **b** $-\dfrac{1}{2}$

34

a $f'(x) = 2x + 8$ **b** 16 **c** $y = 16x - 1$ **d** $-\dfrac{1}{16}$

e $y = -\dfrac{x}{16} + \dfrac{253}{4}$

35

a $y = 32$ **b** $x = 4$

36

a $\dfrac{dy}{dx} = 2x + b$ **b** 4 **c** $\dfrac{63}{4}$ **d** 30

Applications

37

a $f'(x) = -2x$ **b** -6 **c** $y = -6x + 45$ **d** $\dfrac{1}{6}$

e $y = \dfrac{1}{6}x + \dfrac{53}{2}$ **f** $x = \dfrac{15}{2}$ **g** $x = -159$ **h** $\dfrac{333}{2}$ units

i $\dfrac{8991}{4}$ units^2

38

a $\dfrac{dy}{dx} = 3x^2 - 20x + 25$ **b** 13

c $-\dfrac{1}{13}$ **d** $x = 84$

e 252 units^2

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